

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (EC) Examination

November 2012

EC 204.01 Electronics Instruments & Measurement

Date: 30.11.2012, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Define : (i) Accuracy (ii) Precision (iii) Standard Deviation (iv) Arithmetic mean [04]
(b) Discuss methods of signal conditioning. [03]
- Q - 2 (a) Draw the basic block diagram of an oscilloscope and explain the function of each block. [05]
(b) Explain the basic circuit for measurement of frequency showing Gate Control F/F [05]
(c) Give the difference between LCD and LED [04]

OR

- Q - 2 (a) Explain the operation of period measurement with the multiple average mode [05]
(b) Draw the structure of LED and explain its operation. [05]
(c) Which are the techniques used to reduce the input frequency? Explain in brief. [04]
- Q - 3 (a) Explain dual trace oscilloscope. What are the advantages of dual trace over dual beam CRO for multiple trace? [05]
(b) Explain a seven segment LED display. [05]
(c) A set of independent current measurements was taken by six observations and recorded as 12.8mA, 12.2mA, 12.5mA, 13.1mA, 12.9mA and 12.4mA Calculate (i) arithmetic mean (ii) the deviation fro mean. [04]

OR

- Q - 3 (a) Explain the working principle of a Q meter. Also outline the factors that causes errors during a Q measurement [05]
(b) How can L, C and R be measured using a skeleton LCR bridge? [05]
(c) A $4\frac{1}{2}$ digit voltmeter is used for measuring voltage measurements [04]
(i) Find its Resolution.
(ii) How would 12.98V be displayed on 10V range?
(iii) How would 0.6973 be displayed on 1V and 10V ranges?

SECTION - II

- Q - 4 (a) What are limitations of Wheatstone bridge and Maxwell bridge? [04]
(b) Define Transducer. Give the difference between active transducer and passive transducer [03]
- Q - 5 (a) Derive frequency measurement equation for Wien bridge with figure [05]
(b) A balanced AC bridge has following constants. [05]
Arm AB (Z_1): $R = 300 \text{ ohm}$ in series with $C = 0.265 \text{ } \mu\text{F}$
Arm BC (Z_2): $R = 450 \text{ ohm}$
Arm DA (Z_3): $R = 200 \text{ ohm}$ in series with $L = 15.9 \text{ mH}$
Arm CD (Z_4): unknown
The frequency of oscillation is 1 KHz .
Find the constant of arm CD.
(c) Explain standard signal generator with block diagram. [04]

OR

- Q - 5 (a) With neat diagram explain Hay bridge in detail. [05]
(b) Explain spectrum analyzer with block diagram. [05]
(c) A Maxwell bridge is used to measure inductive impedance. The bridge constants at balance are : [04]
 $C_1 = 0.01 \text{ } \mu\text{F}$, $R_1 = 470 \text{ k}\Omega$, $R_2 = 5.1 \text{ k}\Omega$, $R_3 = 100 \text{ k}\Omega$
Find the series equivalent of the unknown impedance.
- Q - 6 (a) Describe construction and operation of Linear Variable Differential Transducer (LVDT) with necessary figures. [08]
(b) What do you mean by strain gauge? Derive an expression for the gage factor in case of bonded wire strain gage. [06]

OR

- Q - 6 (a) Explain Function Generator with neat block diagram. [05]
(b) Explain fundamental suppression harmonic distortion analyzer. [05]
(c) Write a short note on Load Cell [04]
